

SigLIP · arXiv 2303.15343 · Sep 14

Stakeholder

seat 3 · 18+3

Why sigmoid · So400M · chain to BEHAVIOR

Paper club · VLA Architectures · dark academic

Problem · Softmax LIP tax

Huge sync batches

Softmax InfoNCE wants
all-gather + max-subtraction
across the full batch

Costly systems

Memory & chip count
blow up with $|B|^2$
and device sync

Poor small-BS

At $BS \ll 16k$, softmax
underperforms; lore says
“just scale the batch”

Motivation for this club

CLIP-style pretrain = the eyes of modern VLAs.
Better loss + systems → better open baselines for PaliGemma → OpenPI → BEHAVIOR.

Sigmoid vs Softmax · the surgical swap

Spine of this paper: pairwise logistic replaces batch InfoNCE

CLIP Softmax (InfoNCE)

- ▶ Asymmetric $i \rightarrow t + t \rightarrow i$ softmax
- ▶ Needs full-batch normalization
- ▶ All-gather / sync across devices
- ▶ Negatives compete exclusively
- ▶ Weak at small batch sizes
- ▶ Memory $\sim |B|^2$ logits + gather

SigLIP Pairwise Sigmoid

- ▶ Independent $+1 / -1$ per pair
- ▶ No global denom / no all-gather
- ▶ Chunked distributed loss (Fig.1)
- ▶ Learnable $t +$ bias b ($b \approx -10$)
- ▶ Strong at $BS \leq 1k-16k$
- ▶ Saturates $\sim 32k$ (larger can hurt)

Loss sketch · pairwise sigmoid

$$L = -1/|B| \sum_{\{i,j\}} \log \sigma(z_{ij} \cdot (t \cdot x_i^\top y_j + b))$$

$z_{ij} = +1$ match · $z_{ij} = -1$ non-match · $t = \exp(t')$ · $b \approx -10$ init

No softmax denom

Each pair is an independent logistic term — no batch NCE

Bias is load-bearing

Extreme neg:pos ratio needs $b \approx -10$ so early grads stay sane

Chunked systems

Permute text shards; only b_{local}^2 logits in memory (Fig.1)

Method · three recipes

SigLiT

Locked pretrained ViT
Trainable text
LiT image-text data
Up to BS=1M
84.5% INet (g/14)

SigLIP

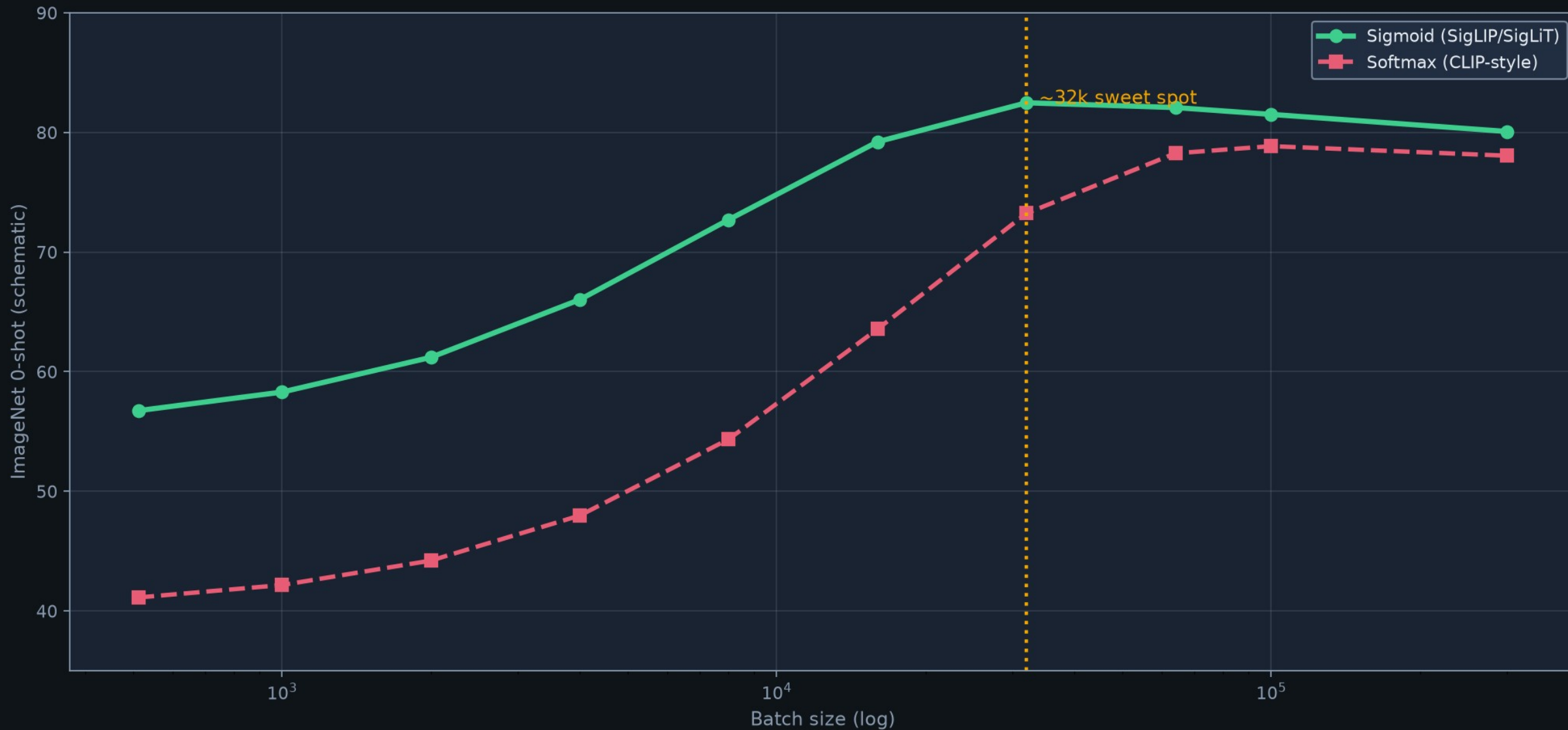
From-scratch ViT+text
WebLI English
Fair CLIP baseline
BS sweet spot 32k
So400M → 83.2%

mSigLIP

B/16 multilingual
Full WebLI 100+ langs
Same ~32k sufficiency
XM3600 retrieval
Vocab memory tricks

Batch story · gains saturate ~32k

Paper Fig. 2 qualitative shape (schematic — cite paper for exact curves)



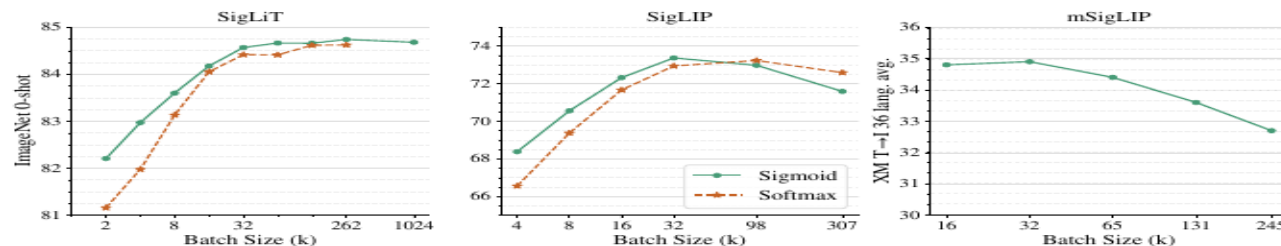


Figure 2: The effect of pre-training batch size. **Left: SigLiT results**, trained for 18B seen examples. Sigmoid loss outperforms the softmax loss significantly with small batch sizes, and performs similarly at larger batch sizes. We successfully trained a SigLiT model with up to *one million* batch size. However, performance for both sigmoid and softmax saturate at around 32 k batch size. **Middle: SigLIP results**, trained for 9B seen examples. Both sigmoid loss and softmax loss saturate at a reasonable batch size, while the peak of the sigmoid loss comes earlier and slightly outperforms the peak of the softmax loss. A very large batch size hurts both losses. **Right: mSigLIP results**, trained for 30B seen examples. With a multilingual setup using over 100 languages, 32 k batch size is surprisingly sufficient and scaling beyond that hurts performance on a 36-language cross-modal retrieval task.

method. In words, we first compute the component of the loss corresponding to the positive pairs, and $b - 1$ negative pairs. We then permute representations across devices, so each device takes negatives from its neighbouring device (next iteration of sum **B**). The loss is then calculated with respect to this chunk (sum **C**). This is done independently in each device, such that each device computes the loss with respect to its local batch b . Losses can then simply be summed across all devices (sum **A**). Individual collective permutes (for sum **B**) are fast (and indeed D collective permutes is typically faster than two all-gathers between D devices), and the memory cost at any given moment is reduced from $|\mathcal{B}|^2$ to b^2 (for sum **C**). Usually b is constant as scaling $|\mathcal{B}|$ is achieved by increasing the number of accelerators. Due to being quadratic with respect to the batch size, the vanilla loss computation rapidly bottlenecks scaling up. This chunked approach enabled training with batch sizes over 1 million on relatively few devices.

4. Results

In this section, we evaluate the proposed SigLiT and SigLIP models across a wide range of batch sizes. We discuss what can be achieved with a small number of accelerator chips, using both SigLiT and SigLIP recipes. We also briefly discuss the impact of batch size on multilingual language image pre-training. We ablate the importance of our large-batch stabilization modification and the introduced learned bias term and present a study on the effect of positive and negative pairs ratio in the sigmoid loss. Lastly,

we explore SigLIP’s data noise robustness.

To validate our models, we report zero-shot transfer results on the ImageNet dataset [14] and zero-shot retrieval results across 36 languages on the XM3600 dataset [44]. We use the ScalingViT-Adafactor optimizer [58] by default for all our experiments.

4.1. SigLiT: Scaling batch size to the limit

Following [59], we use the same precomputed embeddings for the images using a ViT-g vision model, and train a base size text tower from scratch with the same hyperparameters using the LiT image-text dataset [59].

We perform a study over a wide range of batch sizes, from 512 to 1 M , demonstrating the impact of batch size for contrastive learning. Results are presented in Figure 2 (left). When the batch size is smaller than 16 k , sigmoid loss outperforms softmax loss by a large margin. With growing batch sizes, we observe that softmax loss quickly catches up and potentially slightly underperforms sigmoid loss with a large enough batch size. Overall, we recommend using the SigLIP recipe for large batch sizes as well, due to the simplicity, compute savings, and straightforward memory efficient implementation.

There is a consensus that contrastive learning benefits from large batch sizes, while most of the existing studies stop at 64 k batch size [59, 35, 10]. We successfully trained a SigLiT model at one million batch size, to explore the limit of contrastive learning. To our surprise, the performance saturates at 32 k batch size, further scaling up the batch size only gives a minor boost, and the model peaks at

Findings that matter for 2026

Small-BS win

Sigmoid $\gg$ softmax
at BS $< \sim 16k$
(Fig. 2 left/middle)

$\sim 32k$ enough

Larger (e.g. 307k)
can hurt; SigLiT
to 1M adds little

So400M 83.2%

Public efficiency
point $\rightarrow$ open VLA
vision tower

SigLiT speed

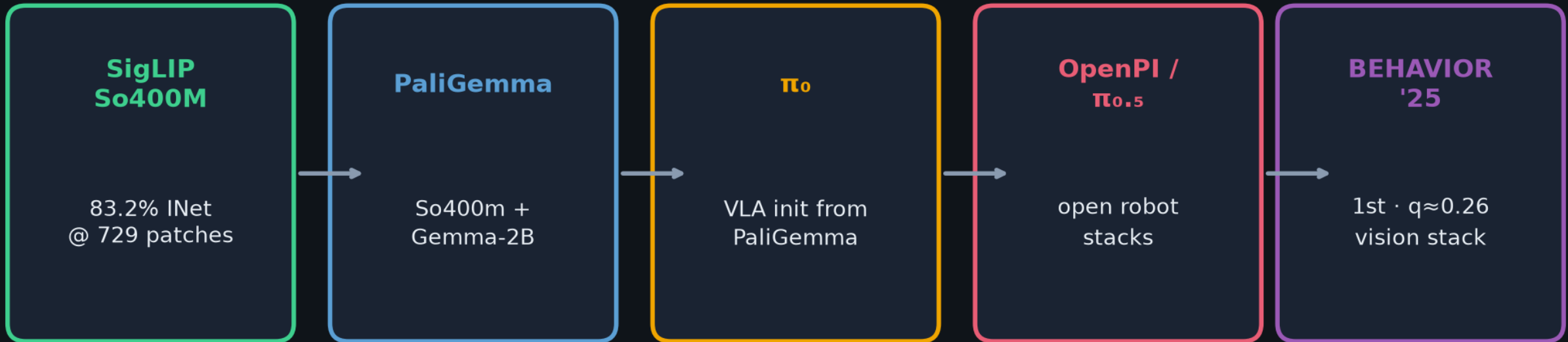
Locked giant vision:
84.5% INet on 4xTPUv4
 ~ 2 days

Lineage payoff

So400m $\rightarrow$ PaliGemma
 $\rightarrow \pi_0$ /OpenPI $\rightarrow$
BEHAVIOR'25 eyes

So400M → open VLA eyes

Public efficiency point that became the default vision tower



Perception backbone — NOT a classical motion planner

Chunking / stages / recovery / data still sit downstream of these eyes

Club thread · long-horizon control

SigLIP IS

- Perception / vision tower
- Default open eyes for VLAs
- Standardized backbone
- Image-text pretrain recipe

SigLIP is NOT

- A motion planner / PDDL
- A substitute for chunking
- A recovery / System-2 loop
- Proof that sigmoid loss = VLA win

BEHAVIOR failures more often sit in chunking / stages / recovery / data

than in ImageNet-0 of the vision tower — but So400m is still the default open eyes those controllers see through.

2026 BEHAVIOR takeaway

Don't reinvent CLIP softmax.

Start from SigLIP / So400M (or SigLIP 2)
and differentiate on

action · data · System-2 · control

Stakeholder close

Don't reinvent CLIP softmax · start from SigLIP/So400M